

Seattle vs. Detroit: Who Rains More? | Weather

The weather project was performed to determine if it rains more in Seattle, WA, or Detroit, MI. For this project, “rains more” is described by precipitation and evaluated on the average precipitation each city experiences overall and how many days each city actually experiences any precipitation.

The datasets for this project were provided by the NOAA (National Oceanic and Atmospheric Administration). A government website that compiles weather data from weather stations around the USA. Two datasets were used from this website, one for Seattle and one for Detroit. These data sets needed to be reorganized and refined to include precipitation measured in inches, over 5 years from 2018 through 2022. This was done using JupyterLab and Python to create a single combined data set containing only the precipitation values for each city and the dates they were measured. To accomplish this, each data set was first inspected to identify the precipitation values for the targeted time period and only included data from one weather station. This inspection also included making sure all the data needed could be accessed and reorganized. For example, the dates included in both data sets needed to be reformatted in a way that allowed the data analysis tools used later to see them. Irrelevant data was then removed from both the Seattle and Detroit data sets respectively. Now that the data was simplified, a follow-up inspection of the data was performed to identify if there were any missing values or data points in each dataset. The Seattle dataset had some missing values that were filled with an estimate. This estimate was calculated using the average precipitation of precipitation measured on the same day as the missing value for all the available years. For example, if there was a missing precipitation value on January 20th, 2018. The estimate used to fill that missing value would come from the average precipitation of January 20th from 2019, 2020, 2021, and 2022. Once completed, the two data sets were merged together, and the resulting combined data set was reformatted and reorganized. Two new columns were added to the data. One calculated the average precipitation each day experienced across the 5 years it was recorded. The second column calculated the number of days each city experienced any precipitation at all. This new clean data set was named “clean_Seattle_Detroit_weather.csv”.

An exploratory data analysis was initiated to find any preliminary impressions of the data. Refer to Figure 1, which depicts all the recorded precipitation values. The initial impression is limited but implies that Detroit saw lower levels of precipitation in 2022 than in the previous recorded years. This is only mentioned as it may indicate a potential limitation in the data if Detroit had experienced an unprecedented weather anomaly that year.

The initial exploratory analysis also includes some basic numerical summaries like the data’s overall average precipitation, standard deviation, minimum precipitation, and max precipitation. These were unremarkable for this dataset, as the average precipitation between the two cities was similar at 0.1003 inches for Detroit and 0.1132 inches for Seattle. The max was also similar at 2.96 inches for Detroit and 2.60 inches for Seattle. Further analyses were conducted, such as the

mean daily precipitation, making a box plot of the general precipitation data, and a bar graph of the overall proportion of days with precipitation. All of which indicated slightly greater precipitation impressions for Seattle. For a deeper analysis of this trend, new columns were created in the data, one calculating the average precipitation per month from the 5-year period and one calculating the proportion of days with precipitation per month from the 5-year period. T-tests and z-tests were conducted on the two new columns, respectively, to determine if any differences between the two cities were statistically significant (labeled with asterisks in the graphs). Figure 2 depicts the average monthly precipitation seen in both cities. Detroit has, on average, more precipitation than Seattle for 4 months in May and late summer are statistically significant. Seattle has, on average, more precipitation than Detroit for 4 months in winter that are statistically significant. This reflects the similarity of the overall precipitation averages between the two cities. Figure 3 depicts the monthly proportion of days with precipitation that each city experienced. This graph clearly indicates that for 6 months, Seattle experiences more days with precipitation, which are statistically significant.

In conclusion, Seattle and Detroit have similar precipitation averages overall, but Seattle experiences a greater proportion of days with precipitation than Detroit. Therefore, Seattle Rains more than Detroit.

Figure 1

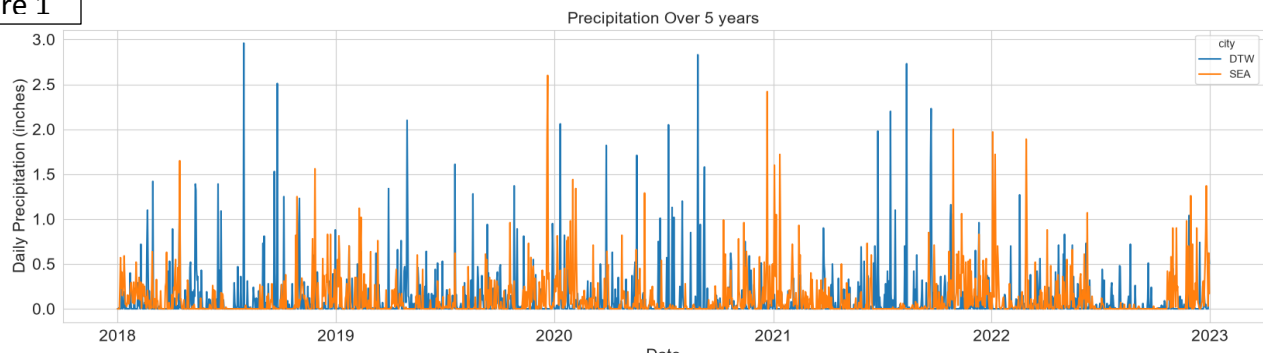


Figure 2

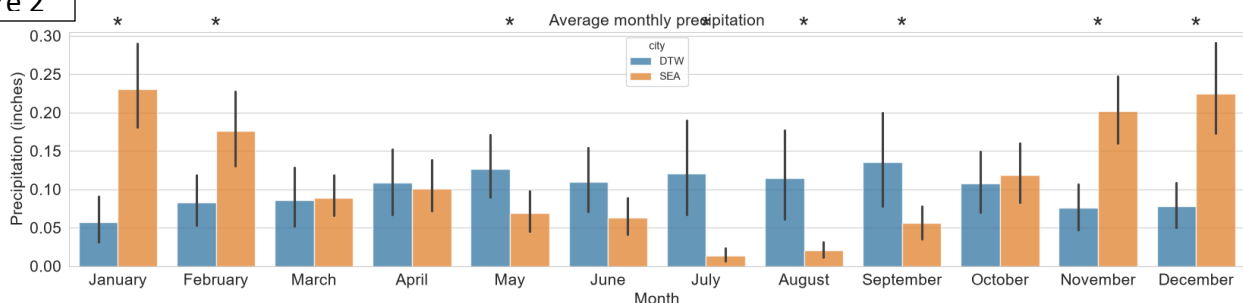


Figure 3

